

AI-Powered Laboratory Assistant

Proposal for BAKER Metrology Lab — MES College of Engineering, Pune

Date: April 12, 2026 **Prepared by:** POC AI Lab Team **Expected Daily Footfall:** 120 students/day
Exchange Rate: ₹93.09 = 1 USD (Morningstar, April 12, 2026, 1:25 PM UTC)

What This System Does

A QR code is placed next to each apparatus in the BAKER Metrology Lab. When a student scans it with their phone camera, they land on a webpage showing photographs of the equipment, a written explanation of the experiment, an audio narration, and a chatbot that answers questions about that specific apparatus.

No app to install. No login needed. Any phone with a camera and a browser works.

On the backend, a private admin panel lets the lab instructor update experiment descriptions, upload new photographs or audio, swap the AI service key, or add new equipment, all without editing any code.

How the Chatbot Works

When a student types a question — say "What is the least count of this micrometer?" — the system sends that question to an AI language model through what's called an API, which is a secure channel that lets our server communicate with the AI provider's servers. The AI reads the question along with pre-loaded context about that experiment, generates an answer, and sends it back. The whole round trip takes about 1–2 seconds.

The AI does not store or remember student conversations. Each question is independent. No personal data is collected or transmitted.

What is a "Token"?

AI providers bill based on **tokens**, not words. A token is roughly three-quarters of a word in English. A 100-word question works out to about 133 tokens; a 200-word answer is about 267 tokens. Every cost figure in this document is calculated in tokens because that's how all three major AI providers invoice.

Current Technical Setup (April 12, 2026)

Component	Technology	Role
Frontend	HTML, CSS, JavaScript	The web pages students see — experiment info, photos, chat
Backend	Python (FastAPI)	The server that handles requests and talks to the AI
AI Model	Google Gemini 2.0 Flash	The language model that generates chatbot answers
Hosting	Vercel (free tier)	Where the website and server code run (cloud-hosted)
Experiments	6 BAKER Lab apparatus	Currently configured and live

Six experiments are configured. The system runs on mobile and desktop browsers. It is live and working today.

⚠ Urgent: The Current AI Model is Being Shut Down

Google has deprecated Gemini 2.0 Flash. It will stop working entirely on **June 1, 2026** — about 7 weeks from today. After that date, every chatbot request will fail.

The older Gemini 1.5 models (1.5 Flash, 1.5 Flash-8B, 1.5 Pro) have already been shut down and return errors if called.

We need to migrate to a newer model before June 1. The replacement options, costs, and our recommendation are detailed below. The migration itself is a one-line configuration change in the code — it does not require rebuilding the system.

Traffic and Usage Estimates

The lab sees approximately 120 students per day. Not every student will use the chatbot. Many will just read the experiment page or listen to the audio and leave. We've modelled two scenarios:

Scenario	Visitors/Day	Chatbot Users	Messages/Day	Monthly Tokens (Input / Output)
Normal	120	54 students (45%)	216	3.9M input / 1.6M output
Worst Case	120	120 students (100%)	1,200	28.8M input / 18M output

Normal assumes 45% of visitors type something into the chatbot, averaging 4 messages each. This reflects a regular teaching week.

Worst case assumes every visitor uses the chatbot heavily. Exam week, workshop days, batch practicals where the whole class needs answers at once.

These numbers drive every cost estimate in this document.

AI Model Pricing — All Major Providers (April 2026)

Every AI company charges per million tokens processed. Below is a comparison across the three major providers. Only currently available models are listed — deprecated and retired models have been removed. All INR figures use ₹93.09 = \$1.

Google Gemini

Google currently offers three generations of models. The **2.5 family** is stable and production-ready. The **3.x family** is in preview (newer, but may change). The **2.0 models** still work today but will be shut down on June 1, 2026.

Model	Input (\$/M)	Input (₹/M)	Output (\$/M)	Output (₹/M)	Free Tier	Status
Gemini 2.5 Flash-Lite	\$0.10	₹9.31	\$0.40	₹37.24	Yes (~1,000 RPD)	✔ Stable
Gemini 2.5 Flash	\$0.30	₹27.93	\$2.50	₹232.73	Yes (~500 RPD)	✔ Stable
Gemini 2.5 Pro	\$1.25	₹116.36	\$10.00	₹930.90	Yes (~100 RPD)	✔ Stable
Gemini 3.1 Flash-Lite Preview	\$0.25	₹23.27	\$1.50	₹139.64	Yes	● Preview
Gemini 3 Flash Preview	\$0.50	₹46.55	\$3.00	₹279.27	Yes	● Preview
Gemini 3.1 Pro Preview	\$2.00	₹186.18	\$12.00	₹1,117.08	No	● Preview
Gemini 2.0 Flash ← current	\$0.10	₹9.31	\$0.40	₹37.24	Yes	⚠ Dies June 1, 2026

RPD = Requests Per Day on the free tier. These are approximate, community-reported limits — Google no longer publishes a static table and instead asks developers to check live limits in

their AI Studio dashboard.

Note on free tier data usage: Google states that queries made on the free tier may be used to improve their products. Paid tier queries are not used this way.

OpenAI (ChatGPT family)

OpenAI's newest release is the **GPT-5.4 family** (March 2026). The GPT-4.1 series remains fully available. GPT-4o is now labelled "legacy" by OpenAI, meaning it works but is no longer their recommended option.

Model	Input (\$/M)	Input (₹/M)	Output (\$/M)	Output (₹/M)	Notes
GPT-4.1 Nano	\$0.10	₹9.31	\$0.40	₹37.24	Cheapest OpenAI model
GPT-4o Mini	\$0.15	₹13.96	\$0.60	₹55.85	Legacy, still works
GPT-5.4 Nano	\$0.20	₹18.62	\$1.25	₹116.36	New generation budget option
GPT-4.1 Mini	\$0.40	₹37.24	\$1.60	₹148.94	Good balance of quality and cost
GPT-5.4 Mini	\$0.75	₹69.82	\$4.50	₹418.91	Mid-range, newer generation
o4-mini	\$1.10	₹102.40	\$4.40	₹409.60	Reasoning model, slower responses
GPT-4.1	\$2.00	₹186.18	\$8.00	₹744.72	Previous flagship, 1M context
o3	\$2.00	₹186.18	\$8.00	₹744.72	Reasoning model, 200K context
GPT-4o	\$2.50	₹232.73	\$10.00	₹930.90	Legacy flagship
GPT-5.4	\$2.50	₹232.73	\$15.00	₹1,396.35	Current flagship, 1.05M context
o1	\$15.00	₹1,396.35	\$60.00	₹5,585.40	Deep reasoning, very expensive
o3-Pro	\$20.00	₹1,861.80	\$80.00	₹7,447.20	Premium reasoning

About reasoning models (o-series): These models "think" internally before answering. That internal thinking consumes output tokens, so a short visible answer can use 5-10 times more tokens than it appears. This makes them significantly more expensive per question than the sticker price suggests.

Anthropic (Claude family)

Anthropic's current lineup is the **Claude 4.6 generation**. Several older models have been retired. Claude 3 Haiku retires on April 20, 2026 (8 days from today). Claude 3.5 Sonnet, Claude 3.5 Haiku, and Claude 3 Opus are already gone.

Model	Input (\$/M)	Input (₹/M)	Output (\$/M)	Output (₹/M)	Context	Notes
Claude Haiku 4.5	\$1.00	₹93.09	\$5.00	₹465.45	200K	Fast, affordable
Claude Sonnet 4.6	\$3.00	₹279.27	\$15.00	₹1,396.35	1M	Strong mid-range
Claude Opus 4.6	\$5.00	₹465.45	\$25.00	₹2,327.25	1M	Most capable

Price drop worth noting: Claude Opus 4.6 costs \$5/\$25 per million tokens. The previous Opus generation (now retired) cost \$15/\$75 — a 67% reduction. Anthropic also offers prompt caching (90% savings on repeated context) and a batch API (50% off) that can stack for up to 95% combined savings on high-volume use.

Monthly AI Cost — Model by Model

These figures translate the per-token rates above into actual monthly bills based on our traffic estimates.

Normal Case (~3.9M input / 1.6M output tokens per month)

Provider & Model	Monthly (USD)	Monthly (₹)	Notes
Gemini 2.5 Flash-Lite (free tier)	\$0.00	₹0	Within ~1,000 RPD free limit at 216 msg/day
Gemini 2.5 Flash-Lite (paid)	\$1.03	₹96	If free quota is exceeded

Provider & Model	Monthly (USD)	Monthly (₹)	Notes
GPT-4.1 Nano	\$1.03	₹96	Same price point as Flash-Lite
GPT-5.4 Nano	\$2.78	₹259	Newer gen, slightly pricier
Gemini 3.1 Flash-Lite Preview	\$3.38	₹314	
GPT-4.1 Mini	\$4.12	₹383	
Gemini 2.5 Flash	\$5.17	₹481	Best upgrade for answer quality
Gemini 3 Flash Preview	\$6.75	₹628	
GPT-5.4 Mini	\$10.13	₹943	
o4-mini	\$11.33	₹1,054	Reasoning model
Claude Haiku 4.5	\$11.90	₹1,108	
GPT-4.1	\$20.60	₹1,917	
o3	\$20.60	₹1,917	Reasoning model
Gemini 2.5 Pro	\$20.88	₹1,943	
Gemini 3.1 Pro Preview	\$27.00	₹2,513	
GPT-5.4	\$33.75	₹3,141	
Claude Sonnet 4.6	\$35.70	₹3,322	
Claude Opus 4.6	\$59.50	₹5,539	
o1	\$154.50	₹14,382	Not practical for this use case

Worst Case (~28.8M input / 18M output tokens per month)

Provider & Model	Monthly (USD)	Monthly (₹)
Gemini 2.5 Flash-Lite	\$10.08	₹938
GPT-4.1 Nano	\$10.08	₹938
GPT-5.4 Nano	\$28.26	₹2,630

Provider & Model	Monthly (USD)	Monthly (₹)
Gemini 3.1 Flash-Lite Preview	\$34.20	₹3,184
GPT-4.1 Mini	\$40.32	₹3,753
Gemini 2.5 Flash	\$53.64	₹4,993
Gemini 3 Flash Preview	\$68.40	₹6,365
GPT-5.4 Mini	\$102.60	₹9,551
o4-mini	\$110.88	₹10,322
Claude Haiku 4.5	\$118.80	₹11,059
GPT-4.1	\$201.60	₹18,767
o3	\$201.60	₹18,767
Gemini 2.5 Pro	\$216.00	₹20,107
GPT-5.4	\$342.00	₹31,837
Claude Sonnet 4.6	\$356.40	₹33,167
Claude Opus 4.6	\$594.00	₹55,295
o1	\$1,512.00	₹1,40,753

Hosting — Where the Website Runs

The website and backend server need to be hosted somewhere on the internet so students can access it. "Hosting" means renting a small slice of computing power in a data centre. Here are the options:

Platform	Monthly (USD)	Monthly (₹)	What You Get
Vercel Hobby	\$0	₹0	What we're using now. Handles 120 visitors/day without issues.
Digital Ocean Basic	\$5	₹465	Simple pricing, good if we move off Vercel
Railway Hobby	\$5	₹465	Keeps the server running continuously
Render Starter	\$7	₹652	Always-on server (their free tier pauses after inactivity, meaning students wait 30+ seconds for a cold start)
AWS Lambda	~\$0.10	~₹9	Charges only when someone visits. Complex to configure.
AWS EC2 t3.micro	\$8.47	₹788	A full virtual machine we control completely
Google Cloud Run	\$0	₹0	2 million free requests/month. Good alternative if already using Google services.
Azure Container Apps	~\$0-\$3	₹0-₹279	Microsoft's option. Free tier available.
Vercel Pro	\$20	₹1,862	Same as Hobby but multiple team members can deploy and manage

Our recommendation: Stay on **Vercel Hobby (₹0/month)**. Our traffic is well within its limits.

Free Supporting Services (₹0 each)

These cloud services handle specific jobs the main hosting doesn't cover:

Service	What It Does	Free Tier Limit
MongoDB Atlas Free	A database that stores experiment data so it survives server restarts	512 MB storage
Cloudinary Free	Stores and delivers audio files and images quickly to students' phones	25 GB bandwidth
Sentry Free	Watches for errors and sends alerts if something breaks	5,000 error events/month
UptimeRobot Free	Checks every 5 minutes if the website is up, sends an email if it isn't	50 monitors

Three Budget Options

Option A: Zero Budget — ₹0/year

Component	Monthly (USD)	Monthly (₹)
Hosting (Vercel Hobby)	\$0	₹0
AI (Gemini 2.5 Flash-Lite, free tier)	\$0	₹0
Database (MongoDB Atlas)	\$0	₹0
Monitoring & CDN	\$0	₹0
Monthly Total	\$0	₹0
Annual Total		₹0

How this works: Gemini 2.5 Flash-Lite has a free tier that allows approximately 1,000 requests per day (with a rate limit of about 15 requests per minute). Normal daily usage is 216 messages, so we'd use roughly 22% of the daily free quota. Even on a heavy workshop day with triple the normal traffic (~650 messages), we'd stay within limits.

If something extreme happens and we exceed the free tier, the overflow is billed at ₹9.31 per million input tokens and ₹37.24 per million output tokens. A full worst-case month costs about ₹938 (\$10.08).

One thing to know: Google states that queries on the free tier may be used to improve their products. For a lab assistant answering metrology questions, this is unlikely to be a concern, but it's worth being aware of.

This is the right choice for a trial semester — prove the system works, collect real usage data, and present results to decision-makers.

Option B: Full Academic Year — ~₹2,343/month normal | ~₹6,855/month worst case

Component	Monthly (USD)	Monthly (₹)
Hosting (Vercel Pro)	\$20.00	₹1,862
AI (Gemini 2.5 Flash, normal)	\$5.17	₹481
AI (Gemini 2.5 Flash, worst case)	\$53.64	₹4,993
Database + Monitoring	\$0	₹0
Monthly Total (normal)	~\$25.17	~₹2,343
Monthly Total (worst)	~\$73.64	~₹6,855
Annual Total (normal)		~₹28,100
Annual Total (worst)		~₹82,260

What changes here: The AI model upgrades to **Gemini 2.5 Flash**, which gives noticeably better answers for technical subjects. When a student asks about measurement uncertainty or how an autocollimator works, 2.5 Flash is more accurate and more detailed than the budget Flash-Lite model. It also has a 1-million-token context window, meaning it can hold much more experiment context in a single conversation. Hosting upgrades to Vercel Pro so that multiple faculty members or TAs can manage the deployment, rather than depending on one person.

The normal-case cost works out to about ₹28,100 per year. The worst case (every student chatting heavily during exam months) pushes it to about ₹82,260. In practice, a mix of normal months and a few heavy months will land somewhere in between.

This is the right tier once the pilot semester proves successful and a modest annual budget is approved.

Option C: Multi-Department Scale — ~₹11,500/month normal | ~₹28,400/month worst case

Component	Monthly (USD)	Monthly (₹)
Hosting (Vercel Pro)	\$20.00	₹1,862
AI (GPT-4.1, normal)	\$20.60	₹1,917
AI (GPT-4.1, worst case)	\$201.60	₹18,767
MongoDB Atlas M10	\$57.00	₹5,306
Sentry Team	\$26.00	₹2,420
Monthly Total (normal)	~\$123.60	~₹11,505
Monthly Total (worst)	~\$304.60	~₹28,355
Annual Total (normal)		~₹1,38,060
Annual Total (worst)		~₹3,40,260

What changes here: The AI model moves to **OpenAI's GPT-4.1**, which handles multilingual questions better (useful if students type in Marathi or Hindi) and has a 1-million-token context window (roughly 750,000 words in a single conversation). The database moves to a dedicated server instead of the free shared tier, meaning faster response times and automatic backups. Error monitoring moves to a paid tier with detailed alert history.

This tier makes sense only if the system expands beyond BAKER to cover 10+ labs across multiple departments.

Side-by-Side Comparison

Option	Monthly — Normal	Monthly — Worst	Annual — Normal	Annual — Worst
A (Zero)	₹0	~₹938	₹0	~₹11,250
B (Year)	~₹2,343	~₹6,855	~₹28,100	~₹82,260
C (Scale)	~₹11,505	~₹28,355	~₹1,38,060	~₹3,40,260

What Needs to Be Fixed Before a Full Launch

The system works today, but five issues need to be addressed before it's ready for a full semester. The first item is time-sensitive.

#	Issue	What Goes Wrong If We Ignore It	Fix	Cost	Deadline
0	Gemini 2.0 Flash is being shut down	The AI chatbot will stop working entirely after June 1, 2026. Every request will return an error.	Switch the API configuration to Gemini 2.5 Flash-Lite (for zero budget) or Gemini 2.5 Flash (if budget is available). This is a one-line change in the server configuration.	₹0	Before June 1
1	Admin changes don't survive restarts	Every time the hosting platform redeploys (which happens automatically during updates), any experiment descriptions the instructor changed are lost. They'd have to re-enter them each time.	Connect a MongoDB Atlas database (free tier) so changes are saved permanently.	₹0	Before launch
2	No rate limiting on the chatbot	Nothing stops a single user from sending hundreds of requests in a minute. One student spamming the chat (or running a script) could burn through the day's free AI quota, locking everyone else out.	Add a server-side limit of 10 requests per minute per device. Students won't notice the limit during normal use.	₹0	Before launch
3	No error alerts	If the AI service goes down (Google has occasional outages), or our server code crashes, nobody finds out until students start complaining. By then, half the day's lab sessions may have had a broken chatbot.	Connect Sentry (free tier) for automatic error alerts. The maintainer gets an email within minutes of any failure.	₹0	Before launch
4	Audio files stored inside the code repository	The audio narration files are saved alongside the source code. This slows down every deployment, increases download times, and bloats the code history over time.	Move audio files to Cloudinary (free CDN), a service designed to deliver media files quickly to users.	₹0	Before launch

Estimated effort: Items 1–4 need about 2 days of development work. Item 0 (the model migration) takes under an hour. No additional budget is needed for any of these — they all use free-tier services.

Our Recommendation

Immediate (before June 1, 2026): Migrate from Gemini 2.0 Flash to **Gemini 2.5 Flash-Lite**. Same price tier (\$0.10/\$0.40 per million tokens), same provider (Google), free tier available. The switch is a single configuration change. This keeps the system running at zero cost while we plan next steps.

For the pilot semester: Run on Gemini 2.5 Flash-Lite's free tier with MongoDB Atlas added for data persistence, plus the four fixes listed above. **Annual cost: ₹0**. Collect real usage data over one semester and present the numbers.

For a full academic year (after the pilot): Upgrade the AI to **Gemini 2.5 Flash** and hosting to **Vercel Pro**. Budget **₹28,000 to ₹82,000 for the year** depending on usage intensity. Gemini 2.5 Flash costs about ₹481/month under normal use and gives measurably better answers for metrology and instrumentation topics — larger context window, more detailed technical explanations, better handling of follow-up questions. Because it's from the same provider (Google), the migration is another one-line configuration change.

What we'd steer away from: Reasoning models — OpenAI's o1, o3-Pro, and similar. These are built for tasks that need deep multi-step thinking: complex maths proofs, legal analysis, research synthesis. They spend 5 to 30 seconds "thinking" internally before they produce an answer, and that internal thinking burns output tokens (so a short-looking reply can cost 5–10x what it appears). A student asking "What is the principle of a sine bar?" does not need that. They need a clear, correct answer in 1–2 seconds. These models are the wrong tool for this job.

All pricing verified against official provider pages (ai.google.dev/gemini-api/docs/pricing, platform.openai.com/pricing, platform.claude.com/docs/en/about-claude/pricing) as of April 12, 2026. AI provider rates change periodically — confirm current figures before submitting a formal budget request. Exchange rate: ₹93.09 = \$1 USD (Morningstar, April 12, 2026, 1:25 PM UTC).